

Домашнее задание № 6

1. Создание БД и таблиц

SELECT

```
table_name,
column_name,
data_type,
is_nullable
FROM information_schema.COLUMNS
WHERE table_schema = 'imdb';
```

COLUMNS 1

SELECT table_name, column_name, data_type, is_ | Введите SQL выражение чтобы отфильтровать резул

	A-Z table_name	A-Z column_name	A-Z data_type	A-Z is_nullable
1	actors	id	UInt32	0
2	actors	first_name	String	0
3	actors	last_name	String	0
4	actors	gender	FixedString(1)	0
5	genres	movie_id	UInt32	0
6	genres	genre	String	0
7	movies	id	UInt32	0
8	movies	name	String	0
9	movies	year	UInt32	0
10	movies	rank	Float32	0
11	roles	actor_id	UInt32	0
12	roles	movie_id	UInt32	0
13	roles	role	String	0
14	roles	created_at	DateTime	0

2. Вставка тестовых данных, используя функцию S3

INSERT INTO imdb.actors

```
SELECT *
FROM s3('https://datasets-documentation.s3.eu-west-3.amazonaws.com/imdb/imdb_ajs_actors.tsv.gz',
'TSVWithNames');
```

Статистика 1

Name	Value
Updated Rows	817718
Execute time	4,267s
Start time	Mon Aug 17 20:45:38 GMT+03:00 2026
Finish time	Mon Aug 17 20:45:43 GMT+03:00 2026
Query	INSERT INTO imdb.actors
	SELECT *
	FROM s3('https://datasets-documentation.s3.eu-west-3.amazonaws.com/imdb/imdb_ajs_actors.tsv.gz',
	'TSVWithNames')

```
INSERT INTO imdb.genres
SELECT *
FROM s3('https://datasets-documentation.s3.eu-west-3.amazonaws.com/imdb/imdb_ijs_movies_genres.tsv.gz',
'TSVWithNames');
```

Статистика 1		
Name	Value	
Updated Rows	395119	
Execute time	0,802s	
Start time	Mon Aug 17 20:46:39 GMT+03:00 2026	
Finish time	Mon Aug 17 20:46:40 GMT+03:00 2026	
Query	INSERT INTO imdb.genres	
	SELECT *	
	FROM s3('https://datasets-documentation.s3.eu-west-3.amazonaws.com/imdb/imdb_ijs_movies_genres.tsv.gz',	
	'TSVWithNames')	

```
INSERT INTO imdb.movies
SELECT *
FROM s3('https://datasets-documentation.s3.eu-west-3.amazonaws.com/imdb/imdb_ijs_movies.tsv.gz',
'TSVWithNames');
```

Статистика 1		
Name	Value	
Updated Rows	388269	
Execute time	2,999s	
Start time	Mon Aug 17 20:47:41 GMT+03:00 2026	
Finish time	Mon Aug 17 20:47:44 GMT+03:00 2026	
Query	INSERT INTO imdb.movies	
	SELECT *	
	FROM s3('https://datasets-documentation.s3.eu-west-3.amazonaws.com/imdb/imdb_ijs_movies.tsv.gz',	
	'TSVWithNames')	

```
INSERT INTO imdb.roles(actor_id, movie_id, role)
SELECT actor_id, movie_id, role
FROM s3('https://datasets-documentation.s3.eu-west-3.amazonaws.com/imdb/imdb_ijs_roles.tsv.gz',
'TSVWithNames');
```

Статистика 1		
Name	Value	
Updated Rows	3431966	
Execute time	10s	
Start time	Mon Aug 17 20:48:17 GMT+03:00 2026	
Finish time	Mon Aug 17 20:48:27 GMT+03:00 2026	
Query	INSERT INTO imdb.roles(actor_id, movie_id, role)	
	SELECT actor_id, movie_id, role	
	FROM s3('https://datasets-documentation.s3.eu-west-3.amazonaws.com/imdb/imdb_ijs_roles.tsv.gz',	
	'TSVWithNames')	

3. Запросы к данным:

а) Найти жанры для каждого фильма

```

SELECT
    m.id AS movie_id,
    m.name AS movie_name,
    m.year,
    m.rank,
    arrayStringConcat(groupArray(g.genre), ', ') AS genres_list
FROM imdb.movies m
INNER JOIN imdb.genres g ON m.id = g.movie_id
GROUP BY m.id, m.name, m.year, m.rank
ORDER BY m.id;

```

movies(+) 1 X

ELECT m.id AS movie_id, m.name AS movie_name | Введите SQL выражение чтобы отфильтровать результаты

	123 movie_id	AZ movie_name	123 year	123 rank	AZ genres_list
1	1	#7 Train: An Immigrant Journey, The	2 000	0	Documentary, Short
2	2	\$	1 971	6,4	Comedy, Crime
3	5	\$1,000 Reward	1 923	0	Western
4	6	\$1,000,000 Duck	1 971	5	Comedy, Family
5	8	\$10,000 Under a Pillow	1 921	0	Animation, Comedy, Short
6	9	\$100,000	1 915	0	Drama
7	10	\$100,000 Pyramid, The	2 001	0	Family
8	11	\$1000 a Touchdown	1 939	6,7	Comedy
9	12	\$20,000 Carat, The	1 913	0	Crime, Drama, Short
10	13	\$21 a Day Once a Month	1 941	0	Animation, Short
11	14	\$2500 Bride, The	1 912	0	Drama, Romance, Short
12	15	\$30	1 999	7,5	Comedy, Short

б) Запросить все фильмы, у которых нет жанра

```

SELECT
    m.id AS movie_id,
    m.name AS movie_name,
    m.year,
    m.rank
FROM imdb.movies m
LEFT JOIN imdb.genres g ON m.id = g.movie_id
WHERE g.movie_id = 0
ORDER BY m.id;

```

movies(+) 1 X

SELECT m.id AS movie_id, m.name AS movie_name | Введите SQL выражение чтобы отфильтровать резул

	123 movie_id	AZ movie_name	123 year	123 rank
1	0	#28	2 002	0
2	3	\$1,000 Reward	1 913	0
3	4	\$1,000 Reward	1 915	0
4	7	\$1,000,000 Reward, The	1 920	0
5	16	\$30,000	1 920	0
6	23	\$50,000 Challenge, The	1 989	0
7	39	'42	1 951	0
8	48	'93 jie tou ba wang	1 993	0
9	49	'94 du bi dao zhi qing	1 994	5,7
10	54	'Abbot' and 'Cresceus' Race, The	1 901	0
11	55	'Akasaka no shimai' yori: yoru no wada	1 960	0
12	63	'Bad Buck' of Santa Ynez	1 915	0

в) Объединить каждую строку из таблицы “Фильмы” с каждой строкой из таблицы “Жанры”

```
SELECT
  m.id AS movie_id,
  m.name AS movie_name,
  m.year,
  m.rank,
  g.genre
FROM imdb.movies m
CROSS JOIN imdb.genres g
;
```

ovies(+) 1 X

ELECT m.id AS movie_id, m.name AS movie_name | Введите SQL выражение чтобы отфильтровать результаты

	123 movie_id	A-Z movie_name	123 year	123 rank	A-Z genre
1	0	#28	2 002	0	Documentary
2	1	#7 Train: An Immigrant Journey, The	2 000	0	Documentary
3	2	\$	1 971	6,4	Documentary
4	3	\$1,000 Reward	1 913	0	Documentary
5	4	\$1,000 Reward	1 915	0	Documentary
6	5	\$1,000 Reward	1 923	0	Documentary
7	6	\$1,000,000 Duck	1 971	5	Documentary
8	7	\$1,000,000 Reward, The	1 920	0	Documentary
9	8	\$10,000 Under a Pillow	1 921	0	Documentary
10	9	\$100,000	1 915	0	Documentary
11	10	\$100,000 Pyramid, The	2 001	0	Documentary
12	11	\$1000 a Touchdown	1 939	6,7	Documentary
13	12	\$20,000 Carat, The	1 913	0	Documentary

г) Найти жанры для каждого фильма, НЕ используя INNER JOIN

```
SELECT
  m.id AS movie_id,
  m.name AS movie_name,
  m.year,
  m.rank,
  (SELECT arrayStringConcat(groupArray(genre), ', ') FROM imdb.genres WHERE movie_id = m.id) AS genres
FROM imdb.movies m
ORDER BY m.id;
```

ovies 1 X

ELECT m.id AS movie_id, m.name AS movie_name | Введите SQL выражение чтобы отфильтровать результаты

	123 movie_id	A-Z movie_name	123 year	123 rank	? genres
1	0	#28	2 002	0	[NULL]
2	1	#7 Train: An Immigrant Journey, The	2 000	0	Documentary, Short
3	2	\$	1 971	6,4	Comedy, Crime
4	3	\$1,000 Reward	1 913	0	[NULL]
5	4	\$1,000 Reward	1 915	0	[NULL]
6	5	\$1,000 Reward	1 923	0	Western
7	6	\$1,000,000 Duck	1 971	5	Comedy, Family
8	7	\$1,000,000 Reward, The	1 920	0	[NULL]
9	8	\$10,000 Under a Pillow	1 921	0	Animation, Comedy, Short
10	9	\$100,000	1 915	0	Drama
11	10	\$100,000 Pyramid, The	2 001	0	Family
12	11	\$1000 a Touchdown	1 939	6,7	Comedy

д) Найти всех актеров и актрис, снявшихся в фильме в N году

```

SELECT DISTINCT
  a.id AS actor_id,
  a.first_name,
  a.last_name,
  a.gender
FROM imdb.actors a
INNER JOIN imdb.roles r ON a.id = r.actor_id
INNER JOIN imdb.movies m ON r.movie_id = m.id
WHERE m.year = 2000
ORDER BY a.last_name, a.first_name;

```

actors(+1) 1 X

SELECT DISTINCT a.id AS actor_id, a.first_name, a.last_name, a.gender

	123 actor_id	A-Z first_name	A-Z last_name	A-Z gender
1	3	Eloy	'Chincheta'	M
2	8	Marcial	'El Jalisco'	M
3	10	Francisco	'El Niño de la Manola'	M
4	12	Antonio	'El Pescaíto'	M
5	15	Antonio	'El Rilete'	M
6	26	Izzy	'The Religious Nut'	M
7	2	Michael	'babepower' Viera	M
8	28	Oswald	'Ô'	M
9	34	Shaggy	2 Dope	M
10	528 791	Diamond	4 Ever	F
11	528 792	Colt	45	F
12	528 795	Michele	A'Court	F
13	60	Satyanarayana	A.	M

Или так

```

SELECT
  a.id AS actor_id,
  a.first_name,
  a.last_name,
  a.gender
FROM imdb.actors a
LEFT SEMI JOIN (
  SELECT r.actor_id
  FROM imdb.roles r
  INNER JOIN imdb.movies m ON r.movie_id = m.id
  WHERE m.year = 2000
) AS actors_in_2000 ON a.id = actors_in_2000.actor_id
ORDER BY a.last_name, a.first_name;

```

actors 1 X

SELECT a.id AS actor_id, a.first_name, a.last_name, a.gender

	123 actor_id	A-Z first_name	A-Z last_name	A-Z gender
1	3	Eloy	'Chincheta'	M
2	8	Marcial	'El Jalisco'	M
3	10	Francisco	'El Niño de la Manola'	M
4	12	Antonio	'El Pescaíto'	M
5	15	Antonio	'El Rilete'	M
6	26	Izzy	'The Religious Nut'	M
7	2	Michael	'babepower' Viera	M
8	28	Oswald	'Ô'	M
9	34	Shaggy	2 Dope	M
10	528 791	Diamond	4 Ever	F
11	528 792	Colt	45	F
12	528 795	Michele	A'Court	F
13	60	Satyanarayana	A.	M

е) Запросить все фильмы, у которых нет жанра, через ANTI JOIN

```
SELECT
  m.id AS movie_id,
  m.name AS movie_name,
  m.year,
  m.rank
FROM imdb.movies m
LEFT ANTI JOIN imdb.genres g ON m.id = g.movie_id
ORDER BY m.id;
```

Результат 1 ✕

SELECT m.id AS movie_id, m.name AS movie_name | Введите SQL выражение чтобы отфильтровать резу

	123 movie_id	A-Z movie_name	123 year	123 rank
1	0	#28	2 002	0
2	3	\$1,000 Reward	1 913	0
3	4	\$1,000 Reward	1 915	0
4	7	\$1,000,000 Reward, The	1 920	0
5	16	\$30,000	1 920	0
6	23	\$50,000 Challenge, The	1 989	0
7	39	'42	1 951	0
8	48	'93 jie tou ba wang	1 993	0
9	49	'94 du bi dao zhi qing	1 994	5,7
10	54	'Abbot' and 'Cresceus' Race, The	1 901	0
11	55	'Akasaka no shimai' yori: yoru no wada	1 960	0
12	63	'Bad Buck' of Santa Ynez	1 915	0
13	65	'Belle de Varsovie, La'	1 993	0